Cepstral Analysis & Mel-Frequency Cepstral Coefficients (MFCC)

by

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Introduction - What is the Cepstrum?

An historical note on Cepstrum and

- Developed while studying echoes in seismic signals (1960s)
- Audio feature of choice for speech recognition / identification
- (1970s) Music processing (2000s)

➢ **Cepstrum** = "spectrum of a spectrum"

➢ Defined as inverse DFT of log-magnitude DFT:

$$\underbrace{c[n]}_{\text{Time-domain signal}} = \underbrace{\mathcal{F}^{-1}\{\underbrace{\log|\overbrace{\mathcal{F}\{x[n]\}}^{\text{Spectrum}}|}_{\text{Log spectrum}}\}}_{\text{Cepstrum}}$$

- Domain: **quefrency** (time-like, measured in seconds)
- Useful for decomposing signals that are **convolved** (source-filter model)

Cepstral Vocabulary

Term	Origin	Meaning
Spectrum	-	FT of signal
Cepstrum	spectrum reversed	Inverse FT of log-spectrum
Frequency	-	variable of spectrum
Quefrency	frequency reversed	variable of cepstrum
Filtering	-	operation on time signal
Liftering	filtering reversed	operation on cepstrum (quefrency domain)
Harmonic	-	multiple of fundamental frequency
Rahmonic	harmonic reversed	multiple of fundamental quefrency
Low-pass lifter	low-pass filter	keeps low quefrencies (filter)
High-pass lifter	high-pass filter	keeps high quefrencies (excitation)

Why Cepstrum for Speech?

Source-filter model

Speech production:

$$s[n] = e[n] * h[n]$$

where $e[n]$ = excitation (voiced: impulse train; unvoiced: noise),
$h[n]$ = vocal tract impulse response.

In frequency domain:

$$S(k) = E(k) \cdot H(k) \qquad \text{where } S(k) = \text{DFT}\{s[n]\} \text{ and so on}$$

We want to separate $e[n]$ and $h[n]$ without knowing them. The cepstrum turns convolution into addition:

$$\log S(k) = \log E(k) + \log H(k)$$

$$c_s[n] = c_e[n] + c_h[n]$$

Homomorphic Deconvolution

Homomorphic filtering steps:

Linear transform (e.g., DTF) $\rightarrow$ convolution becomes multiplication

Logarithm $\rightarrow$ multiplication becomes addition

Inverse linear transform $\rightarrow$ back to time domain (quefrency)

$$s[n] \longrightarrow \mathcal{F} \longrightarrow \log|\cdot| \rightarrow \mathcal{F}^{-1} \longrightarrow c[n]$$

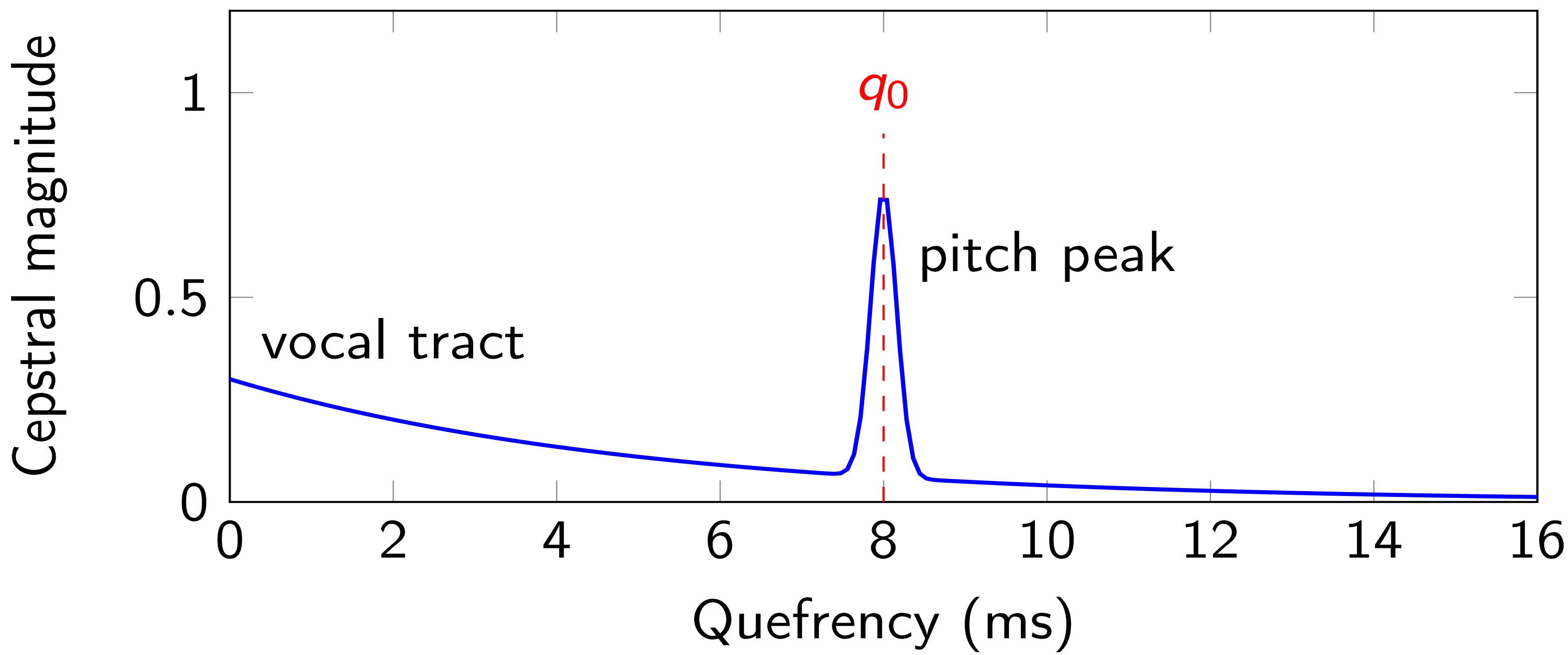
Low quefrency part $\rightarrow$ vocal tract (smooth spectrum)

High quefrency part $\rightarrow$ excitation (pitch harmonics)

Liftering (quefrency-domain filtering) separates them

$\text{Real cepstrum : use only}$ log $|X(k)|$, discard phase $c_r[n] = \mathcal{F}^{-1}\{\log|X(k)|\}$

Illustration - Cepstrum of a Voiced Vowel



q_0 = pitch period (e.g., 8 ms $\rightarrow$ 125 Hz). Unvoiced: no peak.

- Low quefrency (0-2 ms) $\longrightarrow$ vocal tract (filter)
- High quefrency peak $\longrightarrow$ pitch period (source)

Real Cepstrum vs. Complex Cepstrum

For a real speech signal, we usually compute the **real cepstrum**:

$$c[n] = F^{-1}\{\log |X(k)|\}$$

Because in many speech tasks (MFCCs, formant analysis, speaker recognition, etc.):

- the spectral envelope is the important information,
- phase is often less perceptually relevant,
- and the real cepstrum is simpler and more stable.

Complex cepstrum

$$\hat{c}[n] = F^{-1}\{\log X(k)\} \quad \text{with} \quad \log X(k) = \log|X(k)| + \mathrm{j} \arg X(k)$$

- Useful in minimum phase analysis, deconvolution, image synthesis, etc.
- However, most speech-processing applications use real cepstrum

Homomorphic Filtering for Source-Filter Separation

Procedure:

- Compute cepstrum $c[n]$
- Apply lifter (window) in quefrency domain:
  - Low-pass: keep $n < Nc$ → filter component $h[n]$
  - High-pass: keep $n > Nc$ → excitation $e[n]$
- Inverse transform (exponentiate, inverse FFT) to recover time signals

N_c : Cutoff quefrency

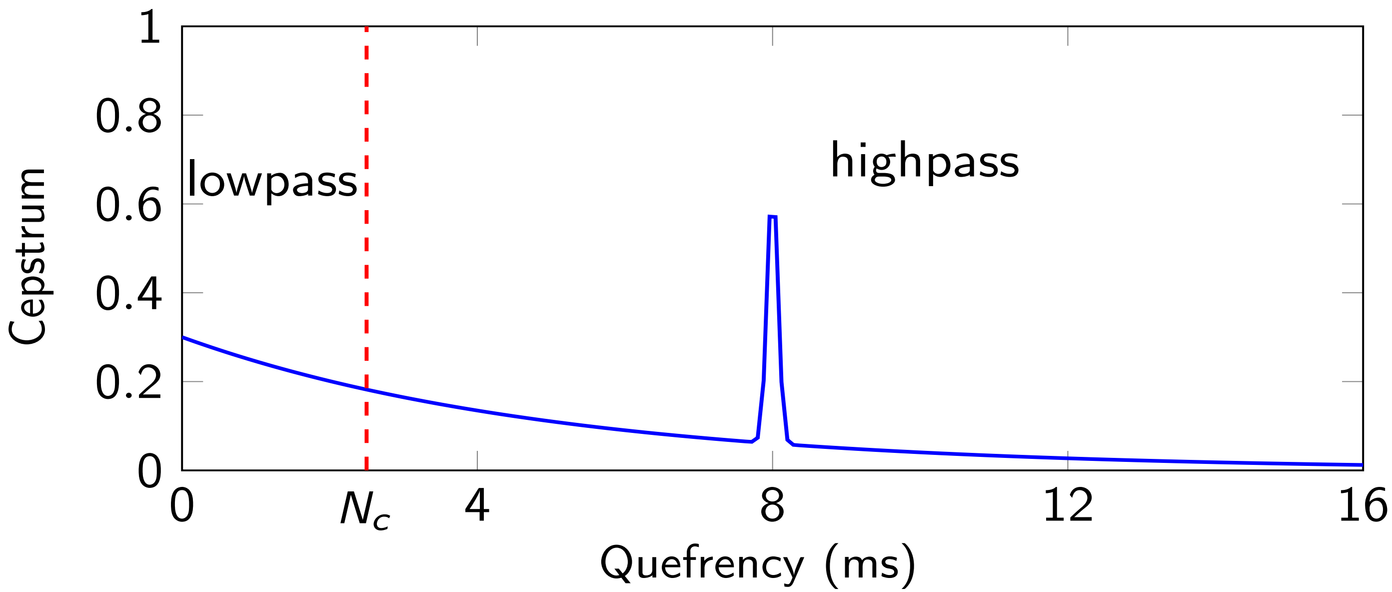
The **cepstral coefficients** for **speech recognition** :

- Keep first K low-quefrency coefficients (e.g., 12-20)

$$\mathbf{c} = [\mathrm{c}_0, \mathrm{c}_1, \mathrm{c}_2, \ldots, \mathrm{c}_{K-1}]$$

- They represent the spectral envelope (vocal tract shape)
- Discard high quefrencies (pitch, noise, speaker-specific details)
- Features d channel variations

Low-Pass Liftering



Cutoff quefrency N_c separates vocal tract (left) from excitation (right).

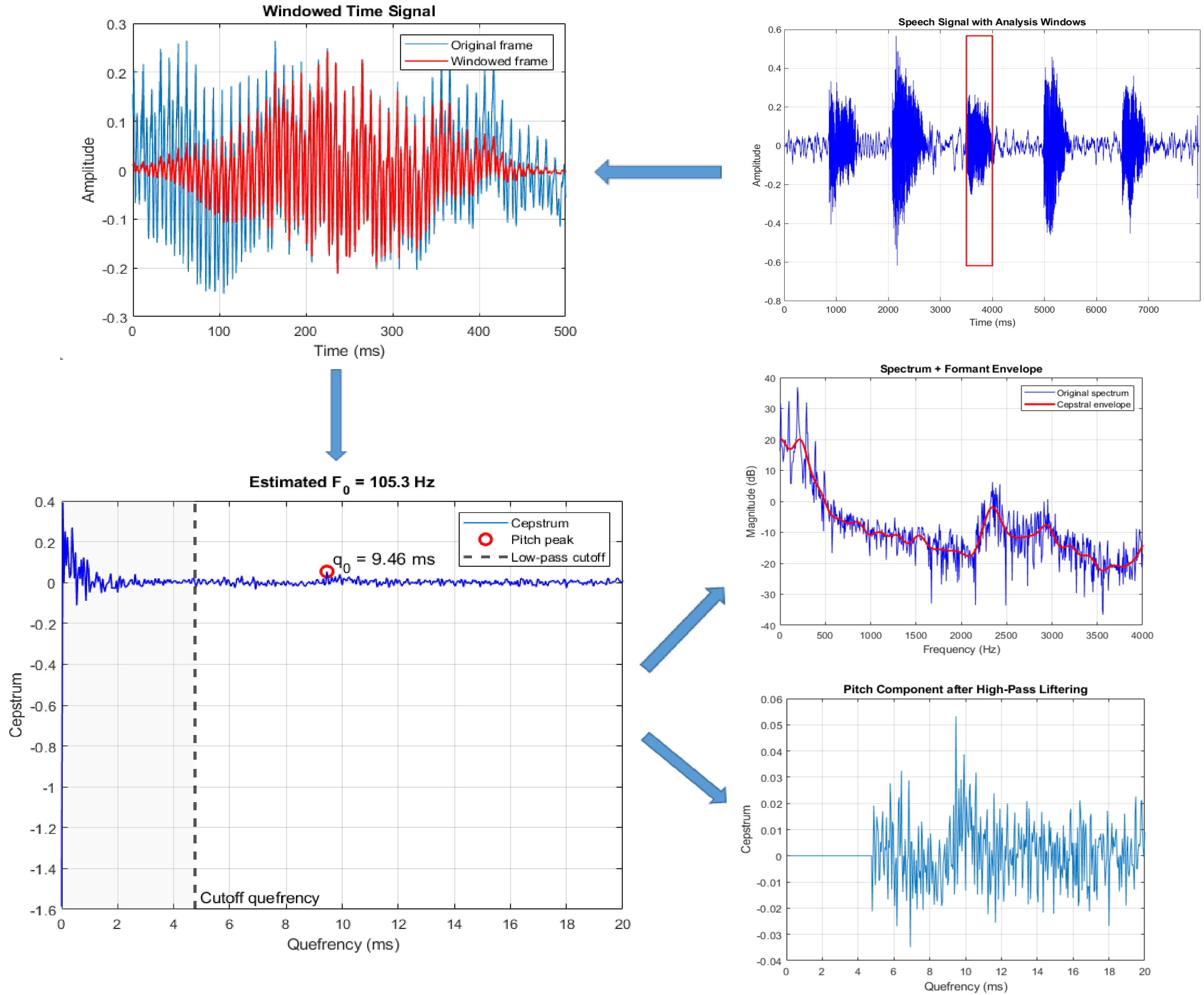
Ideal low-pass lifter: $l_{lp}[n] = \begin{cases} 1 & |n| \leq N_c \\ 0 & \text{otherwise} \end{cases}$

Filtered cepstrum: $c_{filter}[n] = c[n] \cdot l_{lp}[n]$.

Spectral envelope: $H(k) = \exp\left(\mathcal{F}\{c_{filter}[n]\}\right)$

This gives smooth spectrum (formants).

Example of Application

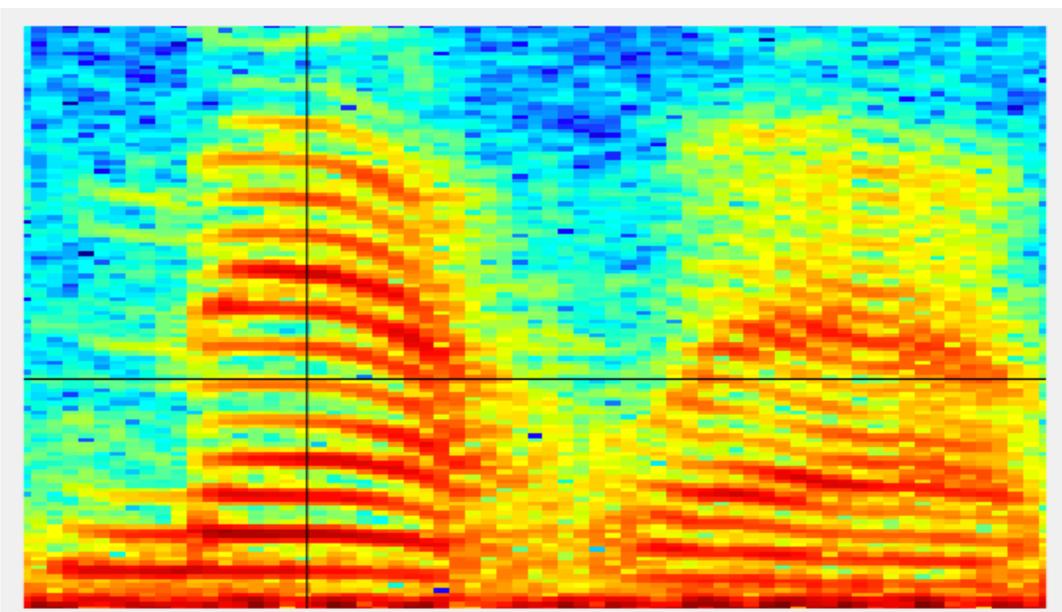


Application - Pitch Detection

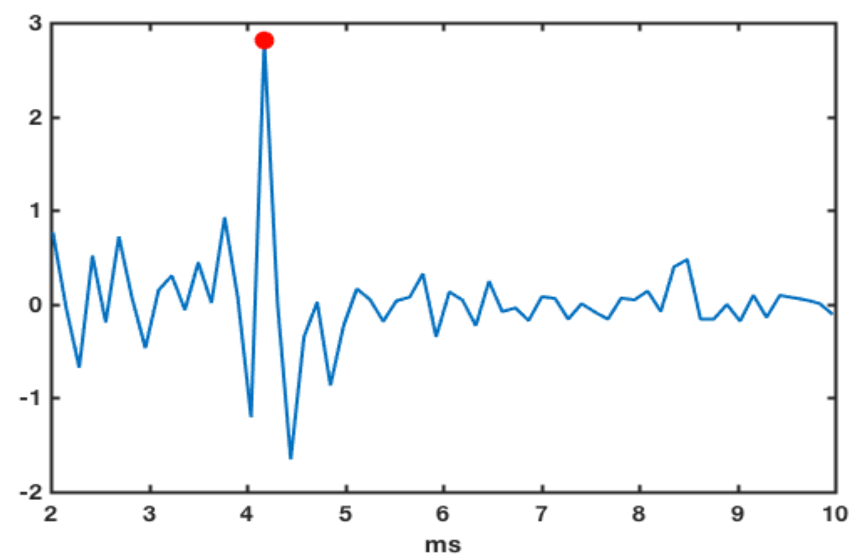
Example: woman saying "Matlab" (real data)

Steps:

- Extract a voiced segment (e.g., vowel /æ/ from 0.1-0.25 s)
- Compute cepstrum
- Search in quefrency range 2-10 ms (100-500 Hz)
- Find maximum peak $\rightarrow$ pitch period
- Convert to frequency: $F_0 = F_s / q_0$



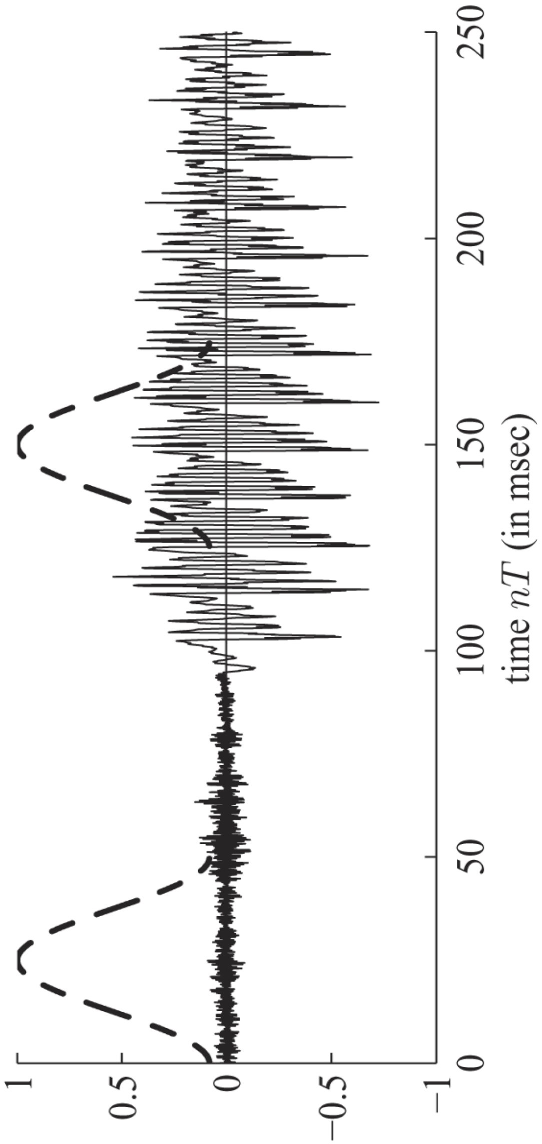
Spectrogram of the word "Matlab" spoken by a woman

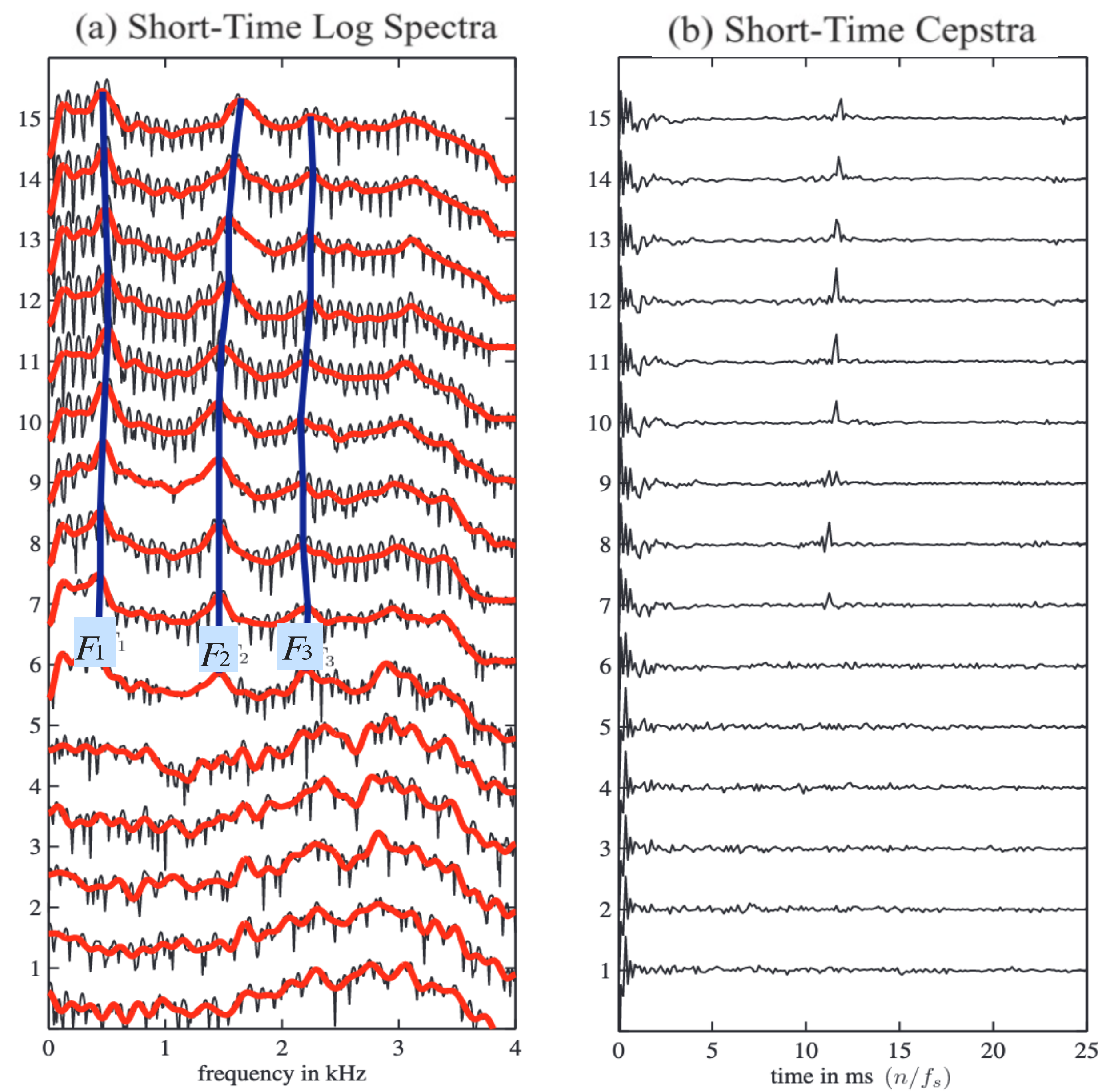


Cepstrum-based estimated F_0 is 239.29 Hz

Cepstral Alanysis: Another example

2001 samples of speech waveform at the transition from the fricative /SH/ to the dipthong /EY/ for an utterance of the word /SH EY D/



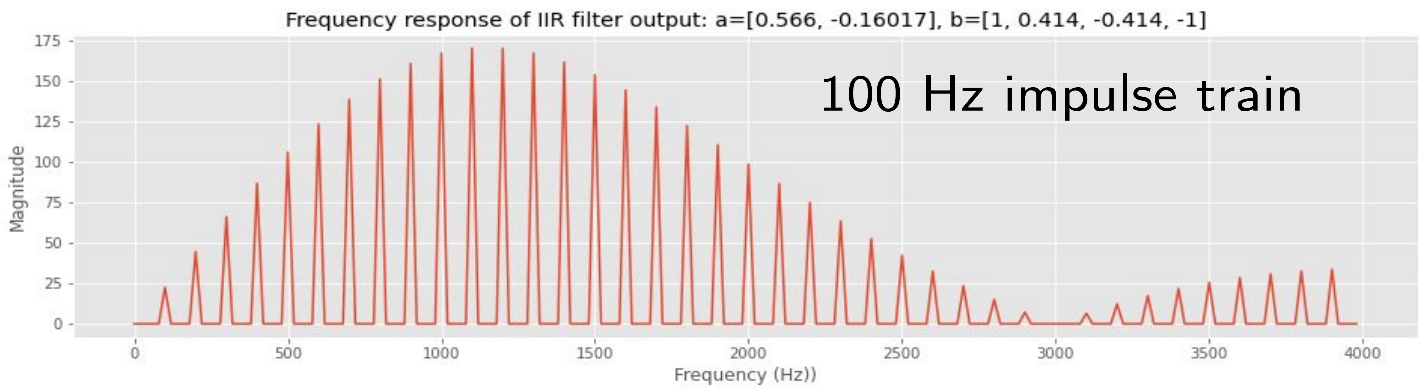


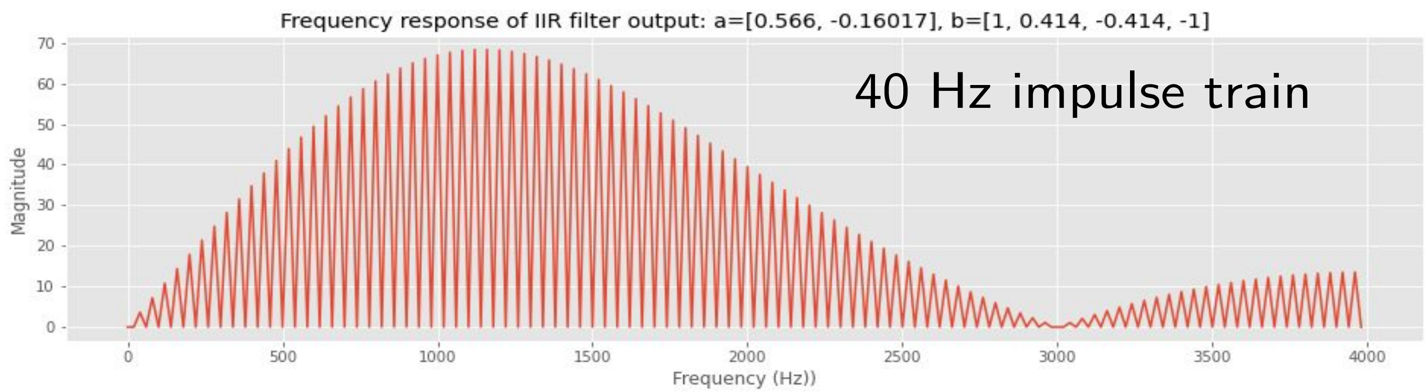
Short-time homomorphic analysis of the waveform in left figure at 15 window positions separated by 100 samples (12.5 msec).

Human speech perception

Recall from the source-filter model:

- Source → F0 and harmonics (e.g. pitch, voicing, loudness)
- Filter → which speech sounds are perceived





- We need the spectral envelope to characterise speech sounds
- We generally don't need the full harmonic structure

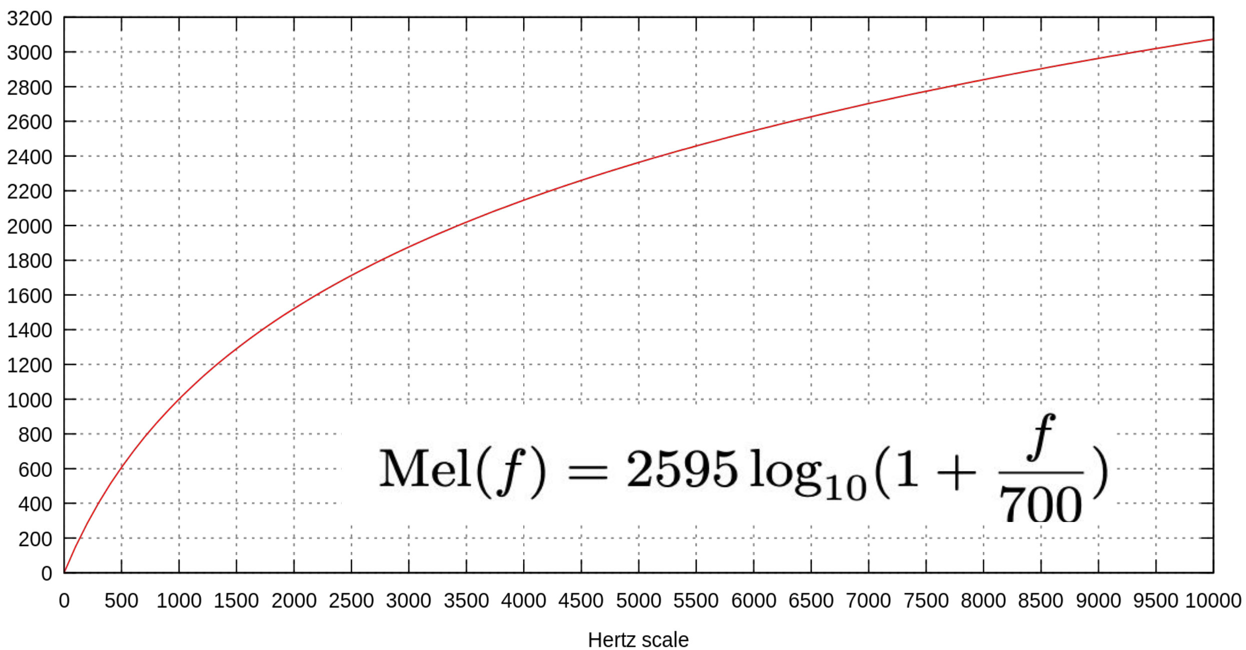
Same filter

Automatic Speech Recognition (ASR) Features

Requirements based on human speech perception:

- Capture the spectral envelope, but abstract away from harmonic structure.
- Human hearing is not equally sensitive to all frequency bands which reflects non-linear human perceptual scales w.r.t frequencies.

Speech Perception - Mel Scale



$$\mathrm{Mel}(f) = 2595 \log_{10}(1 + \frac{f}{700})$$

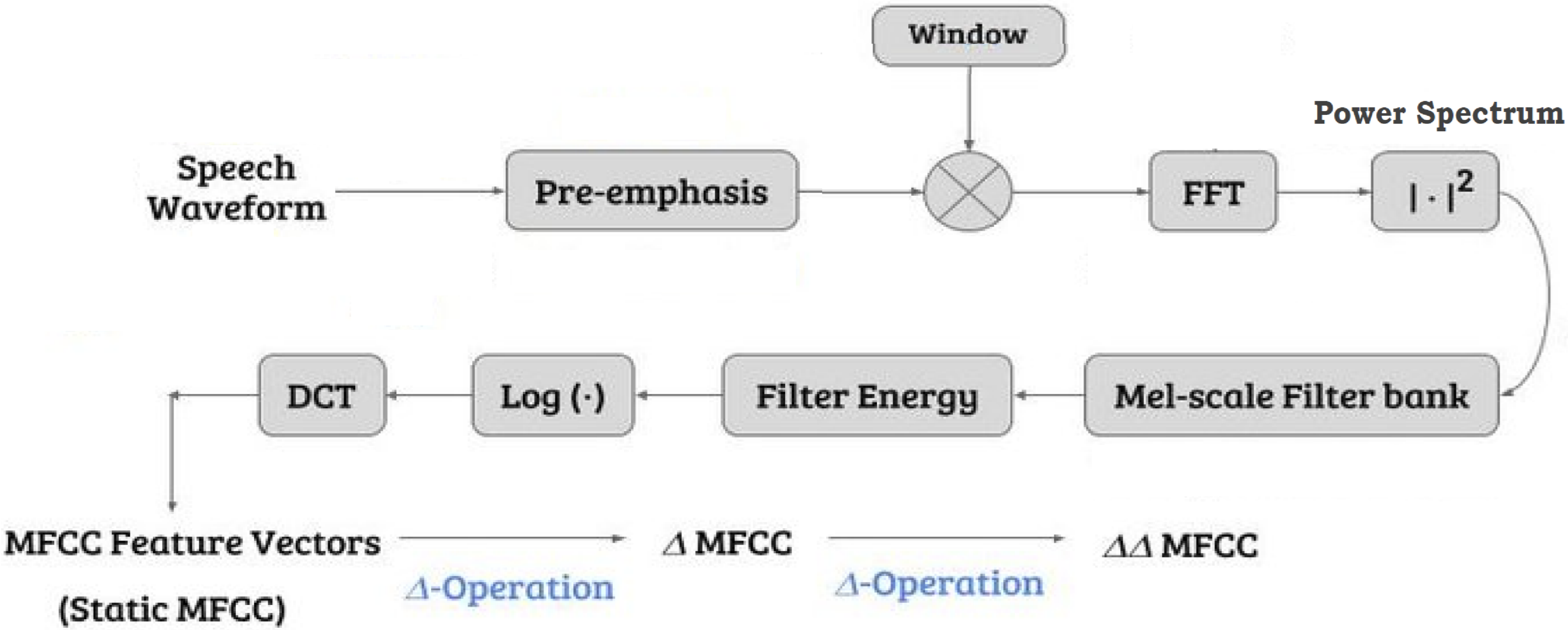
The Mel scale is the "usual" way to map the relationship between frequency (Hz) and perceived pitch (mel)

Mel-scale is approximately linear below 1 kHz and logarithmic above 1 kHz

Mel-Frequency Cepstral Coefficients

- MFCCs are features engineered to fulfill those criteria.
- Basic MFCC extraction steps:
  - Pre-emphasis
  - Windowing
  - DFT $\rightarrow$ magnitude spectrum $\rightarrow$ power spectrum
  - Mel Filterbanks
  - Log
  - Discrete Cosine Transform

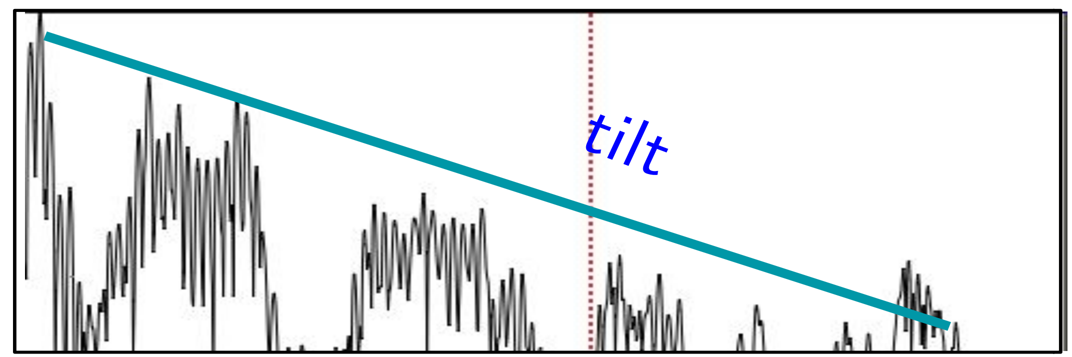
Mel-Frequency Cepstral Coefficients



MFCC - Pre-emphasis

- Higher frequencies in the human voice generally contain less energy. This is called spectral tilt. It is caused by the nature of the glottal pulse.

- We can make the higher frequencies noticeable by applying a filter:



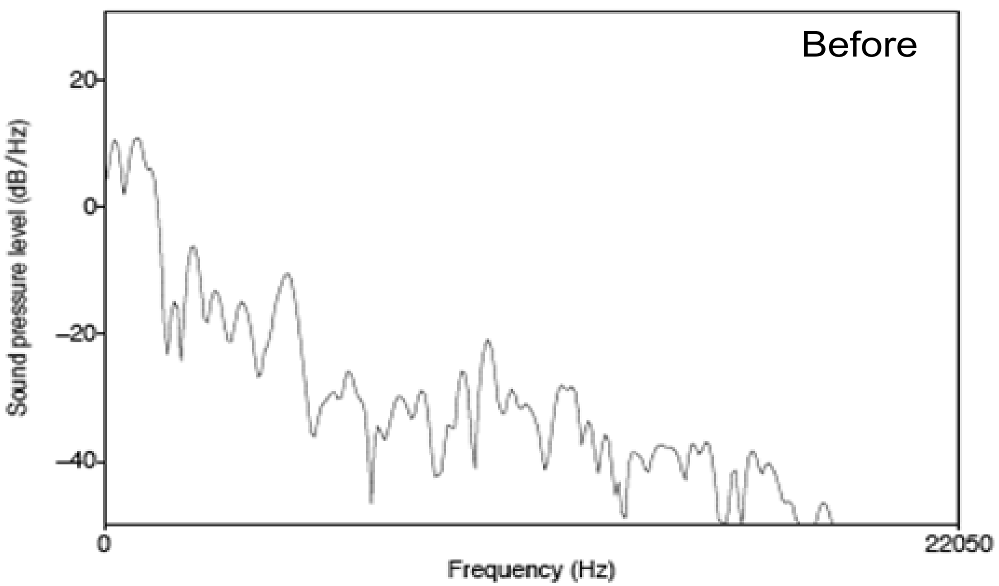
$$x[n] = x'[n] - \alpha\, x'[n-1]$$

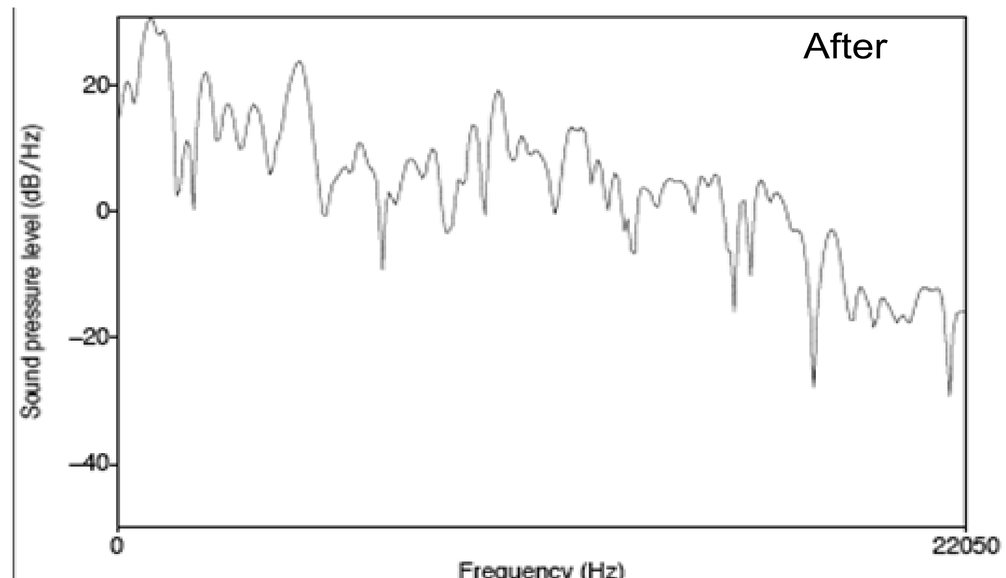
- Boosting high-frequency energy gives more information to tha acoustical model improving thus the speech sound recognition performance

MFCC - Pre-emphasis

Apply filter in time-domain:
$x[n] = x'[n] - 0.97x'[n-1]$

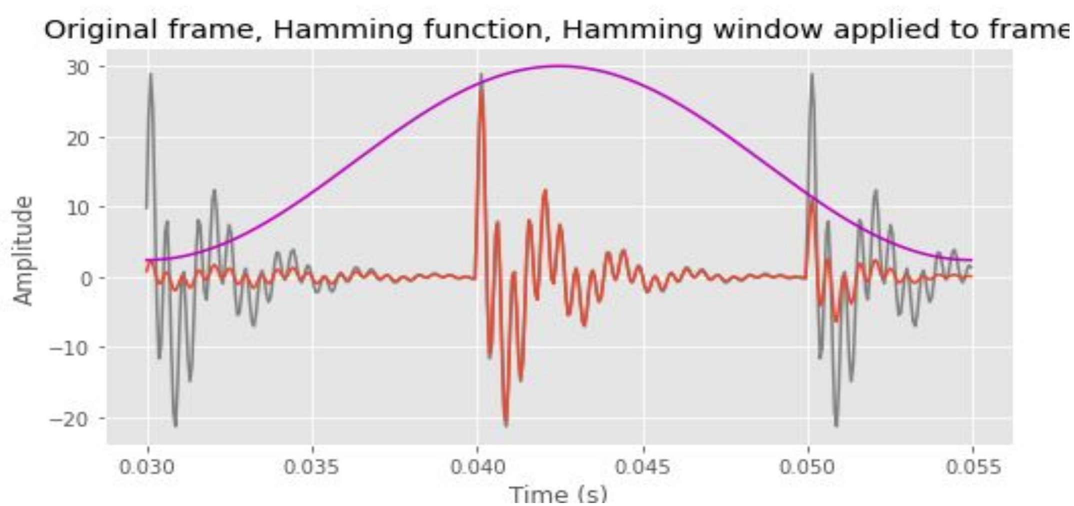
Spectral slice from the vowel [aa] before and after pre-emphasis





MFCC - Windowing

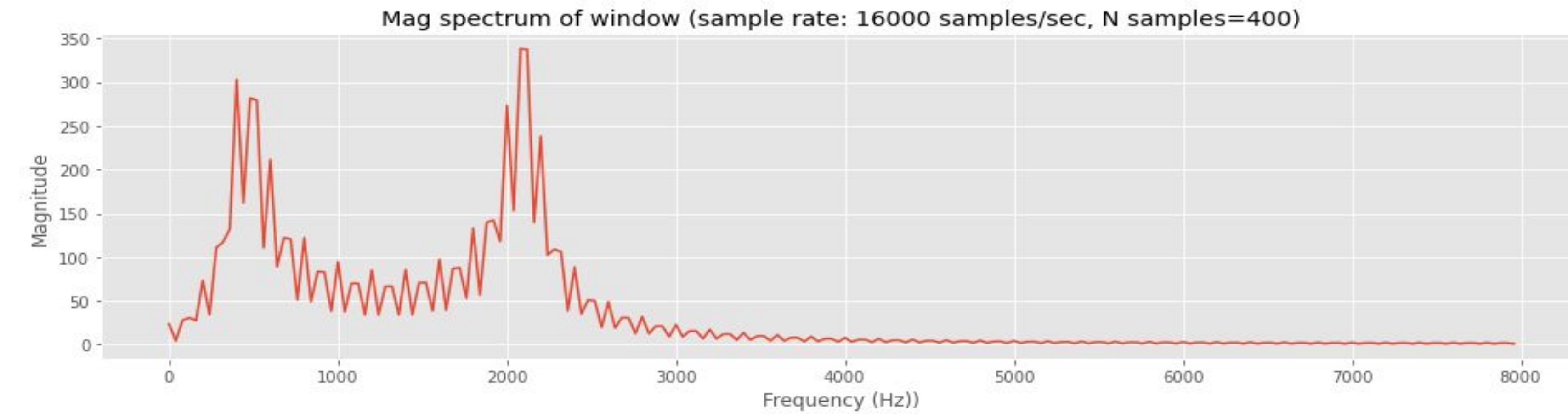
- The DFT assumes stationarity: the same frequency components are present over the entire analysis window
- But we want to know how the spectrum changes over time
- Use a tapered window to extract a many short segments of speech (frames) over time
- Usual frame size: 25ms, frame step 10ms

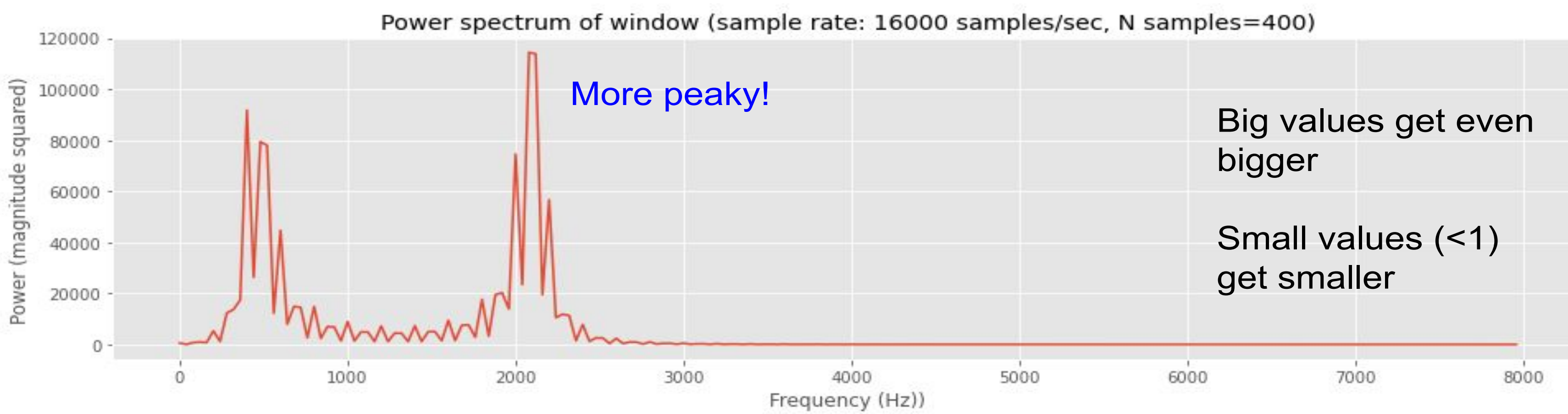


MFCC - DFT: Magnitude and Power spectra

Apply the DFT to each window t, $x_t[n]$:

- The t^{th} DFT output: $X_t[k]$, a complex number
- Spectrum magnitude $= |X_t[k]|$
- Spectrum power $= |X_t[k]|^2$



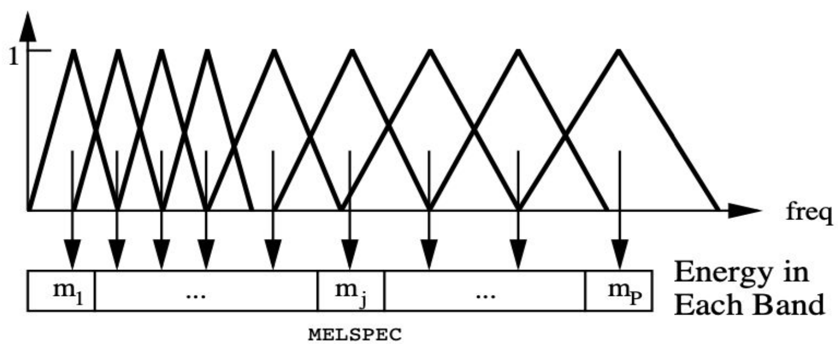


MFCC - Cepstral analysis

We don't want the harmonic structure of the spectrum, just the shape of the spectral envelope

MFCCs:

- Log Mel-filterbank
- Discrete Cosine Transform



Mel-Cepstrum:

- Apply the Inverse DFT to the spectrum directly to capture the shape of the spectrum. (See Part 1 of this course)
- Used more in text-to-speech (TTS) synthesis where we often preserve more spectral details (formants, harmonics, fine spectral shape) to make the synthetic voice sound more natural and intelligible.

MFCC - Mel Filterbank Features

Mel Filter bank: Uniformly spaced before 1 kHz while logarithmic scale after 1 kHz. More resolution in the lower frequencies → capture first few formants.

- We don't want the harmonic structure of the spectrum → take weighted averages over different parts of frequency spectrum (i.e., apply band pass filters)

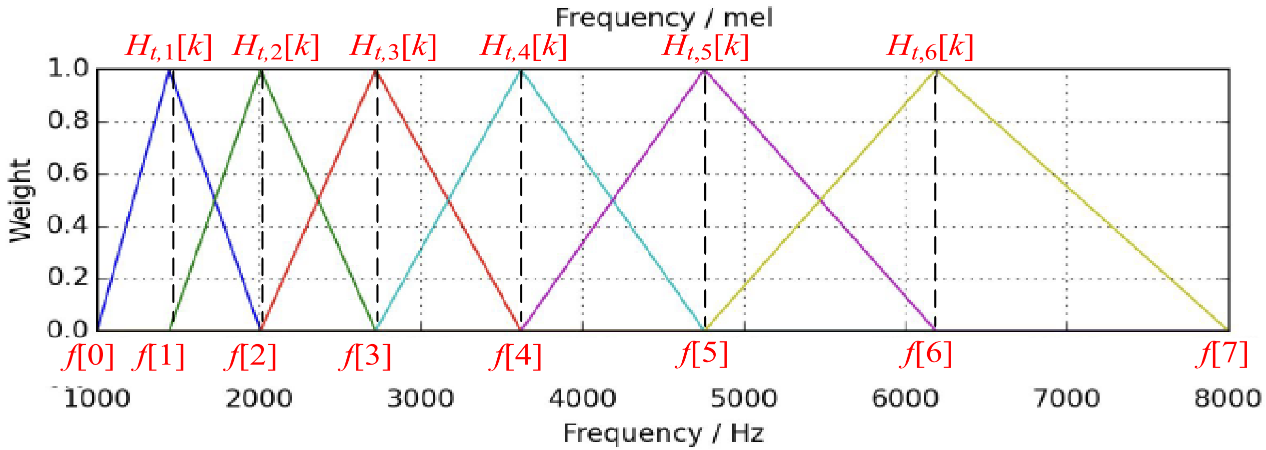
A filterbank with M filters ($m = 1, 2, \dots, M$), where filter m is triangular filter, are defined by:

$$H_{t,m}[k] = \begin{cases} 0 & k < f[m-1] \\ \dfrac{(k - f[m-1])}{(f[m] - f[m-1])} & f[m-1] \le k \le f[m] \\ \dfrac{(f[m+1] - k)}{(f[m+1] - f[m])} & f[m] \le k \le f[m+1] \\ 0 & k > f[m+1] \end{cases}$$

with

$$f[m] = \left(\frac{N}{F_s}\right) B^{-1}\left(B(f_l) + m\frac{B(f_h) - B(f_l)}{M+1}\right)$$

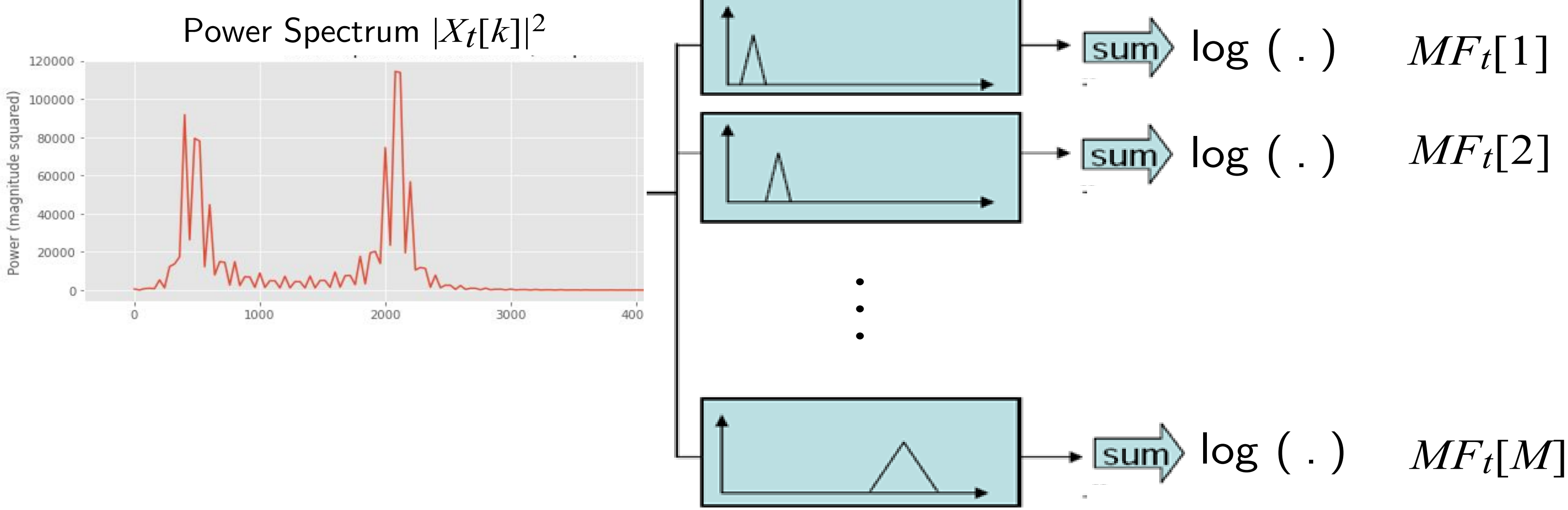
where B and B_{-1} the mel-scale function (given in Slide #14) and its inverse, resp. f_l and f_h are the lowest and highest frequencies of the filterbank in Hz, F_s the sampling frequency and N the size of the FFT.



Each filter bank feature represents the amount of energy in each of these frequency bands

MFCC - Mel Filterbank and log Mel Filterbank

- Apply the bank of filters according Mel scale to the power spectrum
- Each filter output is the sum of its filtered spectral components
- Compute the logarithm of the output



- The log-energy of the m^{th} mel-filter is defined as

$$MF_t[m] = \log_{10} \sum_{k=0}^{N-1} H_{t,m}[k]\ |X[k]|^2 \qquad 1 \leq m \leq M$$

MFCCs and Dynamic MFCCs

- Since the log power spectrum is real and symmetric, inverse DFT reduces to a Discrete Cosine Transform (DCT). The mel frequency cepstrum is then computed as the the DCT of the M filter outputs to form the function $MFCC_m[d]$

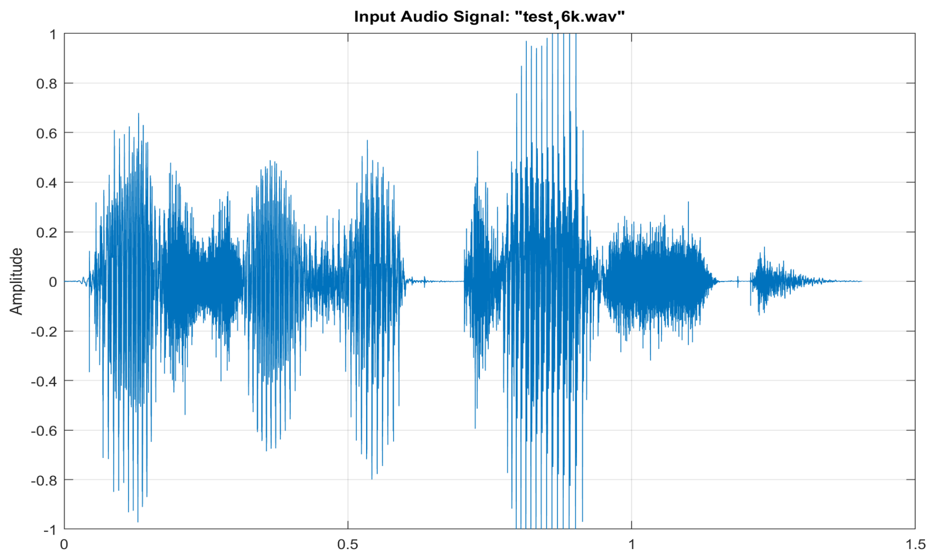
$$MFCC_t[d] = \sum_{m=1}^{M} MF_t[m] \cos(\pi d(m - 1/2)/M)$$

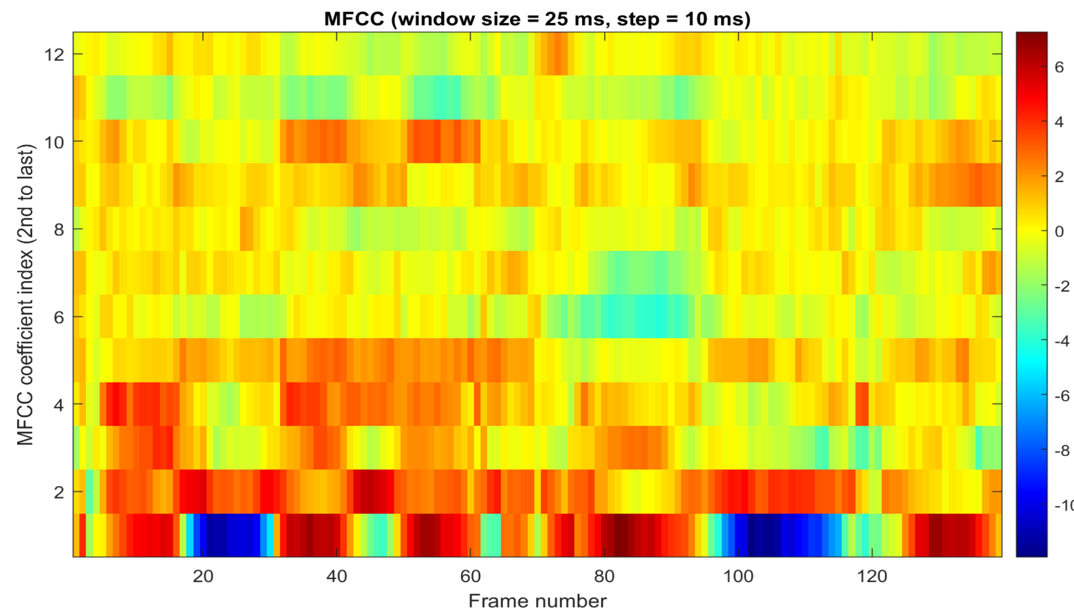
- Typically, $MFCC_m[d]$ is evaluated for $d = 1, 2, \ldots, N_{MFCC}$, where N_{MFCC} is less than the number of mel-filters, e.g., $N_{MFCC} = 12$ and $M = 24$.
- The cepstral coefficients do not capture energy. So we add an energy feature as

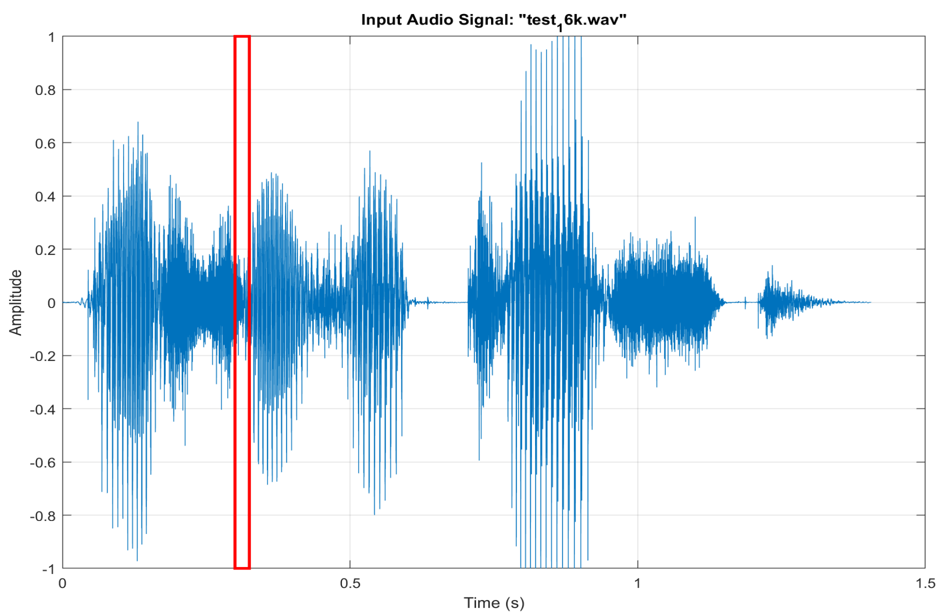
$$\text{logEng} = \log \sum_t |x[k]|^2$$

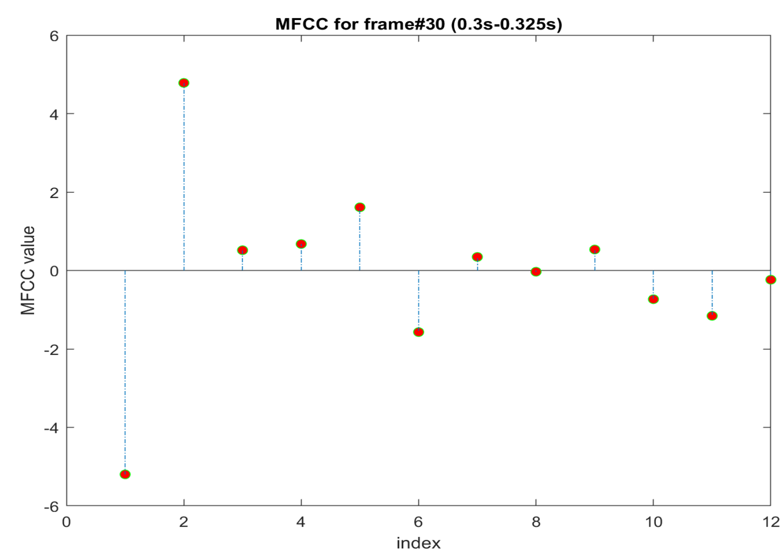
- The set of mel frequency cepstral coefficients provide perceptually meaningful and smooth estimates of speech spectra, over time.
- Since speech is inherently a dynamic signal, it is reasonable to seek a representation that includes some aspect of the dynamic nature of the time derivatives (both first and second order derivatives) of the short-term cepstrum.
- The resulting parameter sets are called the delta cepstrum $\Delta MFCC$ (first derivative) and the delta-delta cepstrum $\Delta\Delta MFCC$ (second derivative).

MFCC - Example of computation of static MFCCs









MFCC - Typical MFCC features

- Window size: 25ms
- Window shift: 10ms
- Pre-emphasis coefficient: 0.97
- Mel Frequency Cepstral Coefficients:
    - 12 MFCC (static)
    - 1 energy feature
    - 12 delta MFCC features
    - 12 double-delta MFCC features
    - 1 delta energy feature
    - 1 double-delta energy feature
- Total **39-dimensional features**

Summary

- Introduced the concept of the cepstrum of a signal, defined as the inverse Fourier transform of the log of the signal spectrum
- Showed cepstrum reflected properties of both the excitation (high quefrency) and the vocal tract (low quefrency)
  - short quefrency window filters out excitation; long quefrency window filters out vocal tract
- Mel-Frequency Cepstral Coefficients (MFCC) used as feature set for speech recognition.
- Delta and delta-delta cepstral coefficients used as indicators of spectral change over time.

References

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